

Still to buy

ZeroDrift Mk3 · Team ZeroDrift, Jabalpur Engineering College · SIH 2026, PS26169 (ISRO)

₹1,747 outstanding across 4 lines — everything else on the ₹6,709 bill of materials is already in hand.
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2× **AS5600 magnetic encoder module** **₹498** (₹249 each)

WHY Closed-loop position feedback. Without these the rig is open-loop and 'accurate' stops being a measurement.

CHECK Small board about 11 × 11 mm with a 4-pin header (VCC, GND, SDA, SCL) and 'AS5600' on the chip.

TRAP Buy TWO. Both are hard-wired to I²C address 0x36, which is why the multiplexer below is not optional.

1× **TCA9548A I²C multiplexer** **₹199**

WHY Both AS5600s answer to 0x36 and cannot share a bus. This gives each one its own channel.

CHECK 8-channel board, roughly 20 × 15 mm, marked TCA9548A or PCA9548A.

TRAP PCA9548A is the same part for our purposes — either is fine.

1× **USB webcam, UVC, 640x480+ — housing removed before mounting** **₹800**

WHY The camera has to ride the gimbal, so a laptop or clip-on webcam that cannot be bolted down is no use. Buy an ordinary UVC webcam and take the housing off.

CHECK The listing must say UVC or 'driver-free / plug and play', and 640×480 or better.

TRAP Do NOT buy an M12 'board camera' or a CCTV module. Those are an industrial category at ₹2,000–5,000 and many are not UVC, so they will not open as a webcam at all. OV7670 and OV2640 modules are parallel DVP — they have no USB and cannot plug into a laptop. The JEC lab has one, but this build needs the HOUSING REMOVED so the bare PCB bolts rigidly inside the head; you likely cannot dismantle issued lab property. Ask first. If the answer is no, buy your own — a camera that is not rigid on the moving head breaks the architecture.

1× **3S 12.6 V balance charger — NOT a single-cell charger** **₹250**

WHY The 18650 pack is three cells in series and there is no mains adapter in this build — the pack is the only motor supply, so it has to be chargeable between runs.

CHECK Listing says 3S and 12.6 V. You plug into the small white JST balance lead, not the main power leads.

TRAP A single-cell 18650 charger cannot charge a 3S pack. Forcing one is a fire risk, not just a slow charge.

Before you glue anything: the diametric magnets must say **DIAMETRIC**, not axial. The two look identical and an axial magnet makes an AS5600 read a constant — both feedback axes dead, with no error message. Test: spin one past a phone compass. The needle must flip twice per rotation. An axial magnet will not.

And when you mount them: each AS5600 chip sits **on the centreline** of the axis it measures, 1.5 mm from the

magnet face. Not near the axis — on it. Off-axis, the chip still answers on I²C and still returns numbers; they are simply not angles.